

SymPy Bug Card

Bug 25: solveset returns a false solution for $\operatorname{acsc}(x) = 3\pi/2$

Task. Solve the inverse cosecant equation over the complex domain.

$$\operatorname{acsc}(x) = \frac{3\pi}{2}$$

SymPy's answer.

$$\operatorname{solveset}\left(\operatorname{acsc}(x) = \frac{3\pi}{2}, x, \mathbb{C}\right) = \{-1\}.$$

Correct answer.

$$\emptyset$$

Explanation. Substitution gives $\operatorname{acsc}(-1) = -\pi/2$, so the returned value has residual -2π . The target $3\pi/2$ is coterminal with $-\pi/2$ for the periodic cosecant function, but it is outside the principal range of acsc .

Likely root cause. The diagnosis locates the bad step in `_invert_complex`, where $\operatorname{acsc}(x) = y$ is inverted mechanically as $x = \operatorname{csc}(y)$ via `acsc.inverse()`, without checking that y is in the principal acsc image. This is a new instance of a known failure mode.

Artifact (GitHub).

[bug-report/bug-025-solveset-returns-a-false-solution-for-acsc-x-3-pi-2/](#)